

APPENDIX 1

Part 5 of 5

Art Unit: 3729

wires (33). (Cf. Abstract). Additionally Mahadevan et al teach that each ground pad of a device is electrically coupled to a ground pad of an adjacent device to form grounded electrical connections or ground wire bonds (32). By aligning the ground wire bonds in the direction that a mold encapsulant will flow during a subsequent molding process the ground wire bonds (32) tend to keep their shapes and not collapse.(Cf. Col. 4). Additionally Mahadevan et al teach a plurality of lags (12-15), a plurality of conductive traces (16, 18, 20 & 22) and a plurality of components (24, 25, 26 and 27) that are electrically connected to form a circuit.(Cf. Col. 3). Additionally Mahadevan et al teach that wire bonding or electrically coupling the semiconductor dies, flags/panels and ground pads the devices are encapsulated. Within a perimeter (34). (Cf. Col. 4). Additionally Mahadevan et al teach that the encasulant can be a solid that is heated to form a liquid and then cooled to form a solid. The encapsulant (35) must enclose at least the grounded wire bond (32) and associated wire device portions (2, 8) since the grounded wire bonds (32) are severed being encapsulated. Additionally Mahadevan et al teach forming a "V" groove that does not extend through the entire lead-frame (10) .(Cf Col. 5). Additionally Mahadevan et al teach cutting (or sawing) device portions with each device portion has a semiconductor die and one ground wire on each side of the semiconductor die. (Cf. Col. 6). Additionally Mahadevan et al teach providing a grounded wire (61) that is electrically coupled to a grounding plane and forms a loop. Second semiconductor (57) is placed of spacer (55) which is placed on semiconductor (54).Semiconductor die (57) is electrically coupled by second wire bond (59). The two semiconductors and wire bonds are encapsulated with the same encasulant that was used for the embodiments showing no stacking of semiconductors (Cf. Col. 8). It would have been obvious to provide a plurality of mold panels, which when assembled would have formed an internal chamber inasmuch as Mahadevan et al teach that the prior art is aware of individually molded packages such as quad-flat-no-lead) components. (Cf. Col 9). As applied to Claims 5 and 6 inasmuch as Mahadevan et al teach cutting (or sawing) portions with each device portion having a semiconductor die the limitation recited in said Claims 5 and 6 are held to have been obvious. As applied to Claims 7-12 a PHOSITA with the used of ordinary skill in this art would treat the exposed contact with a removable conformable coating e.g. a solder or a precious plated metal coating in

Art Unit: 3729

order to prevent oxidation and hence the limitation recited in said Claims 7-12 would have been obvious in view of Mahadevan et al teaching.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CARL ARBES whose telephone number is (571)272-4563. The examiner can normally be reached on M, T, R and F from 8 to 6.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, P Vo, can be reached on (571) 272-4690. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://portal.uspto.gov/external/portal>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/CARL ARBES/

Primary Examiner, Art Unit 3729

Notice of References Cited	Application/Control No. 14/635,420	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner CARL ARBES	Art Unit 3729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-5,557,142 A	09-1996	Gilmore; Peter B.	H01L23/552	174/521
*	B	US-6,734,552 B2	05-2004	Combs; Edward G.	H01L23/4334	257/707
*	C	US-7,015,587 B1	03-2006	Poddar; Anindya	H01L25/0657	257/666
*	D	US-7,030,469 B2	04-2006	Mahadevan; Dave S.	H01L21/561	257/659
*	E	US-7,049,682 B1	05-2006	Mathews; Douglas Jay	H01L23/552	257/659
*	F	US-7,198,987 B1	04-2007	Warren; Robert W.	H01L23/552	257/E23.114
*	G	US-8,138,584 B2	03-2012	Wang; Zhi-jie	H01L21/561	257/659
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0168002	Application No.
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant VLT, Inc.	
		Filing Date	Group Art Unit

U.S. Patent Documents

Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	2002/0096348	07/25/2002	Saxelby et al.			
	2	2004/0100778	05/27/2004	Vinciarelli et al.			
	3	2007/0241440	10/18/2007	Hoang et al.			
	4	2010/0197150	08/2010	Smejtek			
	5	2010/0246152	09/30/2010	Lin et al.			
	6	2012/0287582	11/15/2012	Vinciarelli et al.			
	7	4,366,342	12/1982	Breedlove			
	8	4,551,747	11/1985	Gilbert et al.			
	9	5,241,133	08/1993	Mullen et al.			
	10	5,728,600	03/17/1998	Saxelby, Jr. et al.			
	11	5,864,092	01/1999	Gore et al.			
	12	6,403,009	06/11/2002	Saxelby, Jr. et al.			
	13	6,830,959	12/2004	Estacio			
	14	6,940,013	09/06/2005	Vinciarelli et al.			
	15	7,361,844	04/22/2008	Vinciarelli et al.			
	16	7,972,143	07/2011	Smejtek			
	17	8,427,269	04/23/2013	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications

Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	18	WO1995/027308	10/12/1995	WIPO				

Other Documents (include Author, Title, Date, and Place of Publication)

Examiner Initial	Desig. ID	Document
	19	AFM Microelectronics Inc., "General Purpose Capacitors", www.afmmicroelectronics.com , 13 pages, published on or before July 26, 2010
	20	International Search Report and Written Opinion, PCT/US2012/37495, May 6, 2013, 18 pages.
	21	Kahn, "Technical Information: Multilayer Ceramic Capacitors – Materials and Manufacture", Microelectronics Inc., 9 pages, published on or before July 26, 2010
	22	U.S. Patent Application Serial No. 12/493,773, entitled "Encapsulation Method and Apparatus for Electronic Modules", Application and Preliminary Amendment filed June 29, 2009, 63 pages (24499-0009001)

Examiner Signature <i>/Carl Arbes/</i>	Date Considered 10/13/15
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.A./



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 11/25/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2849

DATE MAILED: 11/25/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: DELIVERING POWER TO SEMICONDUCTOR LOADS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/25/2021

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 11/25/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: DELIVERING POWER TO SEMICONDUCTOR LOADS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/25/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2849	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	11/25/2020	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2849	
DATE MAILED: 11/25/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 10/28/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-113. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date ____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material ____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☒ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/CASSANDRA F COX/
Primary Examiner, Art Unit 2849

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

EXAMINER'S AMENDMENT

1. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

The application has been amended as follows: The abstract has been rewritten as a single paragraph.

--ABSTRACT

Encapsulated electronic modules having complex contact structures may be formed by encapsulating panels containing a substrate comprising pluralities of electronic modules delineated by cut lines and having conductive interconnects buried within terminal holes and other holes drilled in the panel within the boundaries of the cut lines. Slots may be cut in the panel along the cut lines. The interior of the holes, as well as surfaces within the slots and on the surfaces of the panel may be metallized, e.g. by a series of processes including plating. Solder may be dispensed into the holes for surface mounting. Two or more panels may be stacked prior to singulation to form module stacks. Multi-cell converters having a large cell pitch may be combined with an interconnection module to provide vertical power delivery to semiconductor devices through a semiconductor power grid having a small pitch. The converters and interconnection modules may be fabricated in panels and stacked prior to singulation.--

Allowable Subject Matter

1. Claims 1-113 are allowed.
2. The following is an examiner's statement of reasons for allowance: Claims 1-16, 36-43, and 46 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 17-23 are allowed because the closest prior art of record fails to disclose a method of making power converters,

Art Unit: 2849

comprising: selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module in combination with the rest of the limitations of the base claim.

Claims 24-26 are allowed because the closest prior art of record fails to disclose a method of making electronic assemblies, the method comprising: wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern; wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; assembling a first panel comprising a plurality of unsingulated electronic modules; assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and

Art Unit: 2849

the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches in combination with the rest of the limitations of the base claim. Claims 27-29 are allowed because the closest prior art of record fails to disclose a method of forming modular circuit assemblies, the method comprising: selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks in

Art Unit: 2849

combination with the rest of the limitations of the base claim. Claims 30-34, and 100-113 are allowed because the closest prior art of record fails to disclose an apparatus comprising: an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level in combination with the rest of the limitations of the base claim. Claims 35, 44-45, and 47 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second

Art Unit: 2849

pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance in combination with the rest of the limitations of the base claim. Claims 48, 2, 7-8, and 10-14 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of

Art Unit: 2849

converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 49-52, and 100-113 are allowed because the closest prior art of record fails to disclose an apparatus comprising: a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die; an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern

Art Unit: 2849

with alternating polarities configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level in combination with the rest of the limitations of the base claim. Claims 53-66 are allowed because the closest prior art of record fails to disclose apparatus for delivering power vertically to a semiconductor die, the die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint, the apparatus comprising: translation interconnections including X-input terminals configured and arranged to align with and electrically connect to the converter output terminals, a plurality of X-output terminals configured and arranged with alternating polarities to align with respective power input terminals of the semiconductor die, and a plurality of vertical conductors to make electrical connections to the X-output terminals; a capacitive array arranged between the power conversion circuitry and the semiconductor die and electrically connected to the

Art Unit: 2849

translation interconnections to provide energy storage; wherein the translation interconnections are configured to provide electrical and mechanical translation between the converter output terminals and the power input terminals of the semiconductor die; wherein the switching power conversion circuitry, the translation interconnections, and the capacitive array, are each configured and arranged within a respective footprint configured to occupy substantially all of the die footprint, and are together configured and arranged to form a vertical stack configured to occupy substantially all of the die footprint in combination with the rest of the limitations of the base claim. Claims 67, 72-83, 88-89, 91-99 are allowed because the closest prior art of record fails to disclose a vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint; the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising: (a) point-of-load ("POL") switching power conversion circuitry; (b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; and (c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals in combination with the rest of the limitations of the base claim. Claims 68-93 are allowed because the closest prior art of record fails to disclose a vertical power delivery

Art Unit: 2849

apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint; the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising: (a) point-of-load ("POL") switching power conversion circuitry; (b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; (c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals; and (d) a plurality of vertical conductors arranged within the die footprint to connect the interdigitated output terminals to the switching power conversion cells in combination with the rest of the limitations of the base claim.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA F COX whose telephone number is (571)272-1741.

The examiner can normally be reached on M-F 6:00-3:30; off alt Fridays.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lincoln Donovan can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA F COX/
Primary Examiner, Art Unit 2849

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 07 November 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

<i>Issue Classification</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08

NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner) (Date)		113	
/CASSANDRA F COX/ 07 November 2020		O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner) (Date)		30	5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	16	16	40	31	25	46	61	61	76	76	91	91	106	106
2	2	26	17	41	32	47	47	62	62	77	77	92	92	107	107
3	3	27	18	42	33	48	48	63	63	78	78	93	93	108	108
4	4	28	19	43	34	49	49	64	64	79	79	94	94	109	109
5	5	29	20	44	35	50	50	65	65	80	80	95	95	110	110
6	6	30	21	17	36	51	51	66	66	81	81	96	96	111	111
7	7	31	22	18	37	52	52	67	67	82	82	97	97	112	112
8	8	32	23	19	38	53	53	68	68	83	83	98	98	113	113
9	9	33	24	20	39	54	54	69	69	84	84	99	99		
10	10	34	25	21	40	55	55	70	70	85	85	100	100		
11	11	35	26	22	41	56	56	71	71	86	86	101	101		
12	12	36	27	23	42	57	57	72	72	87	87	102	102		
13	13	37	28	24	43	58	58	73	73	88	88	103	103		
14	14	38	29	45	44	59	59	74	74	89	89	104	104		
15	15	39	30	46	45	60	60	75	75	90	90	105	105		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 07 November 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST Search (USPGPUB, USPAT, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	05/26/2020	cfc
H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074 (text only - see search history)	05/26/2020	cfc
inventor search	05/26/2020	cfc
update	07/20/2020	cfc
update	08/17/2020	cfc
update	10/10/2020	cfc
update	11/07/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H02M	1/08, 2001/0074	05/26/2020	cfc
H01L	24/26	05/26/2020	cfc
H05K	1/144, 1/0298	05/26/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
/C.F.C/	1.	10791645	9/29/2020	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
/C.F.C/	2.	Braun et al. "Opportunities of Wafer Level Embedded Technologies for MEMS Devices"; SEMI MEMS Tech Seminar; (possibly published on September 26, 2013); 22 pages
/C.F.C/	3.	"Ideas for attaching / connecting / stacking one PCB onto another with no gap"; http://electronics.stackexchange.com/questions/45733/ideas-for-attaching-connecting-stacking-one-PCB-onto-another-with-no-gap ; accessed on June 3, 2015; 5 pages
/C.F.C/	4.	PCT International Search Report and Written Opinion, PCT/US2012/37495, May 6, 2013, 18 pages
/C.F.C/	5.	USPTO Office Action in U.S. Application No. 13/105,696, dated September 25, 2013, Examiner Arbes, 8 pages
/C.F.C/	6.	USPTO Office Action in U.S. Application No. 14/116,642, dated February 9, 2016, Examiner Shrestha, 15 pages
/C.F.C/	7.	USPTO Office Action in U.S. Application No. 14/635,420, dated October 22, 2015, Examiner Arbes, 6 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 11/12/2020
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2849
Application No. :	16/218,418	Examiner :	Cassandra F. Cox
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed after a request for continued examination and before the mailing of a first action.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: November 10, 2020 _____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Bibliographic Data

Application No: 16/218,418

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: DELIVERING POWER TO SEMICONDUCTOR LOADS

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/12/2018	327	2849	00614-0183001
RULE			

APPLICANTS

VLT, Inc., Andover, MA,

INVENTORS

Patrizio Vinciarelli, Boston, MA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15616288 06/07/2017 PAT 10277105

15616288 is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/04/2019

STATE OR COUNTRY

UNITED STATES

ADDRESS

FISH & RICHARDSON P.C. (VLT, INC)

P.O. BOX 1022

MINNEAPOLIS, MN 55440-1022

UNITED STATES

FILING FEE RECEIVED

\$5,600

Sheet 1 of 2

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VL T, Inc.	
	Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	4326765	4/27/1982	Brancaleone			
	2.	5471366	11/28/1995	Ozawa			
	3.	6147876	11/1/2000	Yamaguchi; Masayoshi			
	4.	6184585	2/6/2001	Martinez et al.			
	5.	6391685	5/21/2002	Hikita et al.			
	6.	6838754	1/4/2005	Kim			
	7.	7298038	11/20/2007	Filoteo et al.			
	8.	8288846	10/16/2012	Narendra et al.			
	9.	8896111	11/1/2014	Vinciarelli			
	10.	9107290	8/11/2015	Chen			
	11.	9571084	2/14/2017	Vinciarelli et al.			
	12.	9584026	2/28/2017	Vinciarelli et al.			
	13.	9660537	5/23/2017	Vinciarelli			
	14.	10701828	6/30/2020	Vinciarelli			
	15.	10757816	8/25/2020	Vinciarelli et al.			
	16.	20030227280	12/11/2003	Vinciarelli et al.			
	17.	20040157410	8/1/2004	Yamaguchi			
	18.	20050167814	8/4/2005	Beroz et al.			
	19.	20070158799	7/1/2007	Chiu; Chin-Tien			
	20.	20080116589	5/22/2008	Li et al.			
	21.	20090251873	10/1/2009	Chenq; Sun-Wen Cyrus			
	22.	20100072604	3/13/2010	Komatsu et al.			
	23.	20100172116	7/1/2010	Yorita			
	24.	20100246152	9/30/2010	Lin et al.			
	25.	20120287582	11/15/2012	Vinciarelli et al.			
	26.	20140152350	6/5/2014	Ikriannikov et al.			

Examiner Signature /CASSANDRA F COX/	Date Considered 11/07/2020
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Substitute Disclosure Form

Sheet 2 of 2

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	27.	20140218155	8/1/2014	Akre			
	28.	20150181719	6/25/2015	Vinciarelli et al.			
	29.	20150181727	6/25/2015	Vinciarelli			
	30.	20160302312	10/13/2016	Vinciarelli et al.			
	31.	20200253060	8/6/2020	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass
	32.	WO 1995/027308	10/12/1995	WIPO			
	33.	WO 2012/155036	11/15/2012	WIPO			

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 11/07/2020
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S239	31	("0000006" "10701828" "10757816" "20030227280" "20040157410" "20050167814" "20070158799" "20080116589" "20090251873" "20100072604" "20100172116" "20100246152" "20120287582" "20140152350" "20140218155" "20150181719" "20150181727" "20160302312" "20200253060" "4326765" "5471366" "6147876" "6391685" "6838754" "7298038" "8288846" "8896111" "9107290" "9571084" "9584026" "9660537").PN.	US-PGPUB; USPAT	OR	ON	2020/11/07 22:07

11/7/2020 11:16:44 PM

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To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Nov 25, 2020 04:09:42 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
16218418	NOA	11/25/2020	00614-0183001
	1449	11/25/2020	00614-0183001
	1449	11/25/2020	00614-0183001

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date
16/218,418	12-12-2018

(Note: The boxes above may be left blank if information is provided on the Transmittal.)

- ☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above:

26214

OR

- ☐ I hereby appoint Practitioner(s) named in the attached list as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above.

Please recognize or change the correspondence address for the application identified in the attached transmittal letter to:

- ☐ The address associated with the above-mentioned Customer Number

OR

- ☒ The address associated with Customer Number: 26214

OR

Firm or Individual Name				
Address				
City		State		Zip
Country				
Telephone		Email		

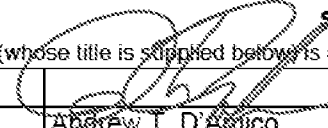
I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

VICOR CORPORATION

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	November 12, 2020
Name	Andrew T. D'Amico		
Title	General Counsel, Intellectual Property		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ *Total of _____ forms are submitted.

Electronic Acknowledgement Receipt

EFS ID:	41308440
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	07-DEC-2020
Filing Date:	12-DEC-2018
Time Stamp:	13:01:59
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	00614_0183001_Supp_ADS.pdf	89615	no	2
			1d0cdb2ac57d1a62b401022a2148bb2ba0518ee9		

Warnings:

Information:					
This is not an USPTO supplied ADS fillable form					
2	Assignee showing of ownership per 37 CFR 3.73	00614_0183001_3_73_Statement.pdf	95582 0acc82a831731f9c3c01381d3f5826e44580ea5c	no	2
Warnings:					
Information:					
3	Req to correct/update/change appl name until pay issue fee	00614_0183001_Req_Correct_Applicant.pdf	127781 2e020eea4b46f6f99bbd59a1c990b7bda17fd42a	no	2
Warnings:					
Information:					
4		00614_0183001_Trans_POA_Vincor.pdf	212287 0dbda3ee79f213614599963733e1a9ffa6b60ea7	yes	2
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	1	
	Power of Attorney		2	2	
Warnings:					
Information:					
Total Files Size (in bytes):			525265		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	<u>16/218,418</u>
Title of Invention	Delivering Power To Semiconductor Loads		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.			

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.	☐ Person who shows sufficient proprietary interest		
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor :			
If the Applicant is an Organization check here. ☒			

Organization Name ~~VELT, Inc.~~ Vicor Corporation

Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1	25 Frontage Road			
Address 2				
City	Andover	State/Province	MA	
Country	US	Postal Code	01810	
Phone Number		Fax Number		
Email Address				

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.				
Signature	/Rex I. Huang/			Date (MM/DD/YYYY)
First Name	Rex	Last	Huang	Registration
				57,661

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	00614-0183001
	Application Number	<u>16/218,418</u>
Title of Invention	Delivering Power To Semiconductor Loads	

		Name		Number	
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24073220.doc

STATEMENT UNDER 37 CFR 3.73(c)Applicant/Patent Owner: VLT, Inc.Application No./Patent No.: 16/218,418 Filed/Issue Date: December 12, 2018Titled: DELIVERING POWER TO SEMICONDUCTOR LOADS

Vicor Corporation, a corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that, for the patent application/patent identified above, it is (choose one of the option 1, 2, 3 or 4 below):

1. ☒ The assignee of the entire right, title, and interest.
2. ☐ An assignee of less than the entire right, title and interest (check applicable box):
- ☐ The extent (by percentage) of its ownership interest is _____. Additional Statement(s) by the owners holding the balance of the interest must be submitted to account for 100% of the ownership interest.
- ☐ There are unspecified percentages of ownership. The other parties, including inventors, who together own the entire right, title and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

3. ☐ The assignee of an undivided interest in the entirety (a complete assignment from one of the joint inventors was made). The other parties, including inventors, who together own the entire right, title, and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

4. ☐ The recipient, via a court proceeding or the like (e.g., bankruptcy, probate), of an undivided interest in the entirety (a complete transfer of ownership interest was made). The certified document(s) showing the transfer is attached.

The interest identified in option 1, 2 or 3 above (not option 4) is evidenced by either (choose one of the options A or B below):

- A. ☐ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel _____, Frame _____, or for which a copy thereof is attached.
- B. ☒ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: Patrizio Vinciarelli To: VLT, Inc.The document was recorded in the United States Patent and Trademark Office at
Reel 048463, Frame 0172, or for which a copy thereof is attached.2. From: VLT, Inc. To: Vicor CorporationThe document was recorded in the United States Patent and Trademark Office at
Reel 054125, Frame 0560, or for which a copy thereof is attached.

[Page 1 of 2]

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(c)

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

4. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

5. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

6. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(c)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (*i.e.*, a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Rex I. Huang/
Signature

December 7, 2020
Date

Rex I. Huang
Printed or Typed Name

57,661
Title or Registration Number

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

**REQUEST TO CORRECT OR UPDATE
THE NAME OF THE APPLICANT UNDER
37 CFR 1.46(c)(1), OR CHANGE THE
APPLICANT UNDER 37 CFR 1.46(c)(2)
(FOR USE ONLY IN APPLICATIONS FILED ON OR AFTER
SEPTEMBER 16, 2012)**

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Art Unit	2849
Examiner Name	Cassandra F. Cox
Practitioner Docket Number	00614-0183001

To: Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Applicant hereby requests that the name of the applicant be corrected or updated under 37 CFR 1.46(c)(1), or that the applicant be changed under 37 CFR 1.46(c)(2), in the above-identified application. Requests under 37 CFR 1.46(c)(1) or (c)(2) cannot be submitted after payment of the issue fee or if the application has been patented.

Please check the applicable box(es) below.

- ☐ 1. This request is to correct or update the name of the applicant (under 37 CFR 1.46(c)(1)) and includes:
- ☐ An application data sheet (ADS) in accordance with 37 CFR 1.76(c) with the corrected or updated information shown with markings (e.g., underlining for insertions, strikethrough for deletions). A Corrected Web-based ADS may be used.
- Note: Request under 37 CFR 1.46(c)(1) may be filed to correct typographical errors in the name of the § 1.46 applicant, or for updating the name of the § 1.46 applicant (i.e., where there is no change in the applicant itself but just in the applicant's name). See the Manual of Patent Examining Procedure (MPEP) section 605.01.
- ☒ 2. This request is to change the applicant (under 37 CFR 1.46(c)(2)) and includes:
- ☒ An application data sheet (ADS) in accordance with 37 CFR 1.76(c) that identifies the changes with proper markings (underlining for insertions and strikethrough for deletions). A Corrected Web-based ADS may be used.
- ☒ A Statement Under 37 CFR 3.73(c) (Form PTO/AIA/96 or equivalent). See MPEP 325.

I am the

☐ applicant* ☒ attorney or agent of record
Registration number 57,661 ☐ attorney or agent acting under 37 CFR 1.34
Registration number _____

Signature /Rex I. Huang/

Type or printed name: Rex I. Huang

Date: December 7, 2020

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. *Juristic entities must be represented by a patent practitioner (See 37 CFR 1.31, applicable to any paper filed on or after September 16, 2012 that is presented on behalf of a juristic entity, regardless of application filing date). Submit multiple forms if more than one signature is required, see below**.

☐ ** Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.36. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 37 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Title	DELIVERING POWER TO SEMICONDUCTOR LOADS
Art Unit	2849
Examiner Name	Cassandra F. Cox
Attorney Docket Number	00614-0183001

SIGNATURE of Applicant or Patent Practitioner

Signature	/Rex I. Huang/	Date (Optional)	
Name	Rex I. Huang	Registration Number	57,661
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ *Total of _____ forms are submitted.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

CONFIRMATION NO. 1092
POA ACCEPTANCE LETTER



Date Mailed: 12/09/2020

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/07/2020.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/agizaw/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/218,418	12/12/2018	2849	24440	00614-0183001	47	6

CONFIRMATION NO. 1092 REPLACEMENT FILING RECEIPT

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022



Date Mailed: 12/09/2020

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Patrizio Vinciarelli, Boston, MA;

Applicant(s)

Vicor Corporation, Andover, MA;

Power of Attorney: The patent practitioners associated with Customer Number 26214

Domestic Priority data as claimed by applicant

This application is a CIP of 15/616,288 06/07/2017 PAT 10277105
which is a CIP of 15/091,346 04/05/2016 PAT 10158357

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/04/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/218,418**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

DELIVERING POWER TO SEMICONDUCTOR LOADS

Preliminary Class

327

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

CONFIRMATION NO. 1092
POA ACCEPTANCE LETTER



Date Mailed: 12/09/2020

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/07/2020.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/mbekele nedy/



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/218,418	12/12/2018	2849	24440	00614-0183001	47	6

CONFIRMATION NO. 1092
CORRECTED FILING RECEIPT

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022



Date Mailed: 12/09/2020

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Inventor(s)

Patrizio Vinciarelli, Boston, MA;

Applicant(s)

Vicor Corporation, Andover, MA;

Power of Attorney: The patent practitioners associated with Customer Number 26214

Domestic Priority data as claimed by applicant

This application is a CIP of 15/616,288 06/07/2017 PAT 10277105
which is a CIP of 15/091,346 04/05/2016 PAT 10158357

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/04/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/218,418**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

DELIVERING POWER TO SEMICONDUCTOR LOADS

Preliminary Class

327

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2849
Application No. :	16/218,418	Examiner :	Cassandra F. Cox
Filed :	December 12, 2018	Confirmation No. :	1092
		Notice of Allowance Date:	November 25, 2020
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP ISSUE FEE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REPLY TO NOTICE OF ALLOWANCE

In response to the Notice of Allowance dated November 25, 2020, enclosed is a completed Part B - Fee(s) Transmittal.

The issue fee is being paid with this reply on the Electronic Filing System. Apply the fee and any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

COMMENTS ON EXAMINER'S REASONS FOR ALLOWANCE

It is recognized that in accordance with M.P.E.P. § 1302.14, the Examiner's reasons for allowance need not set forth all of the details as to why the claims are allowed. In the above-referenced application, it is not conceded that the Examiner's stated reasons for allowance are the only reasons for which the claims are allowable. The Examiner's reasons for allowance indicate that particular claim elements are not disclosed or suggested by the prior art of record, yet the claims may be patentable for other reasons as well, including the inventive combination of all of the recited claim elements. It is not conceded that the specific limitations identified by the Examiner are necessary to distinguish the art of record or to satisfy the requirements of 35 U.S.C. § 112. Moreover, the Examiner does not assert, and it would not be conceded, that the Examiner's reasons have any bearing on the patentability of claims in any other applications directed to the disclosed subject matter.

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 2 of 2

Attorney Docket 00614-0183001 / 0269-US-1

In addition, each dependent claim stands on its own and may be allowable on its own merits. In particular, each dependent claim may be allowable on the basis of a combination of some of the features recited in the dependent claim and its base claim(s), which combination of features may not include all of the limitations identified in the Examiner's reasons for allowance.

Respectfully submitted,

Date: December 8, 2020 _____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

24078175.doc

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22314-1450

By fax, send to: (571) 273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

26214 7590 11/25/2020

FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: DELIVERING POWER TO SEMICONDUCTOR LOADS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	LARGE	\$1200	\$0		\$1200	02/25/2021
EXAMINER		ART UNIT	CLASS-SUBCLASS			
COX, CASSANDRA F.		2849				

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 Fish & Richardson P.C.
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE
Vicor Corporation(B) RESIDENCE: (CITY and STATE OR COUNTRY)
Andover, MAPlease check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☒ Advance Order - # of Copies 5

4b. Method of Payment (Please first reapply any previously paid fee shown above):

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5. Change of Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27.
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid Certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken as a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken as a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Rex I. Huang/Date December 8, 2020Typed or printed name Rex I. HuangRegistration No. 57,661

Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
PRINTED COPY OF PATENT - NO COLOR	8001	5	3	15
Total in USD (\$)				1215

Electronic Acknowledgement Receipt

EFS ID:	41325679
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	08-DEC-2020
Filing Date:	12-DEC-2018
Time Stamp:	15:16:45
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1215
RAM confirmation Number	E2020B8F17021748
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Resp_NOA.pdf	170621	yes	3
			7769774868c603ded04284bf226822a8c9f187ec		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Post Allowance Communication - Incoming		1	2	
	Issue Fee Payment (PTO-85B)		3	3	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	31865	no	2
			32e80d0bc744190869da6a2f14c18259e995a879		
Warnings:					
Information:					
Total Files Size (in bytes):			202486		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

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Preliminary Class

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Dec 09, 2020 03:57:13 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	N570	12/09/2020	00614-0183001
	APP.FILE.REC	12/09/2020	00614-0183001
	N570	12/09/2020	00614-0183001
	APP.FILE.REC	12/09/2020	00614-0183001
	APP.FILE.REC	12/09/2020	00614-0183001

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Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

.S. Patent Documents

Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	7015587	03/21/2006	Poddar			
	2.	7030469	4/18/2006	Mahadevan et al.			
	3.	7198987	4/3/2007	Warren et al.			
	4.	7494843	2/24/2009	Lin et al.			
	5.	8138584	3/20/2012	Wang et al.			
	6.	20070241440	10/18/2007	Hoang et al.			
Change(s) applied to document,	7.	20080078572	7/26/2011	Watanabe et al.	04/2008		
/M.H.E./	8.	20140355218	12/4/2014	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications

Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass
	9.	JPH04293293	10/16/1992	JP	English Abstract		

Other Documents (include Author, Title, Date, and Place of Publication)

Examiner Initial	Desig. ID	Document
------------------	-----------	----------

Examiner Signature /CASSANDRA F COX/	Date Considered 07/20/2020
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

PRINTER RUSH
(PTO ASSISTANCE)

Application: 16218418

Examiner: Cox

GAU: 2849

From:

Location: FDC

Creation Date: 12/15/2020

Week Date: 8/17/2020

<u>DOC CODE</u>	<u>DOC DATE</u>	<u>MISCELLANEOUS</u>
☐ 1449		☐ Continuing Data
☐ IDS		☐ Foreign Priority
☒ CLM	<u>09/25/2020</u>	☐ Document Legibility
☒ IIFW/FWCLM	<u>11/25/2020</u>	☐ Fees
☐ SRFW		☐ Petition (TC)
☐ DRW		☐ Other
☐ OATH		
☐ 312		
☐ SPEC		

[RUSH] Message:

Please correct the claim dependencies

Per the set of claims dated 09/25/2020 and the most recent claim index (11/25/2020), original claims 2, 7, 8 and 10 - 14 all now depend on New claim 48. A claim can not depend on a subsequent numbered claim.

Thank You

[XRUSH] Response:

Initials: _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	12/24/2020		
FISH & RICHARDSON P.C. (VLT, INC)			EXAMINER	
P.O. BOX 1022			COX, CASSANDRA F	
MINNEAPOLIS, MN 55440-1022				
			ART UNIT	PAPER NUMBER
			2849	
			NOTIFICATION DATE	DELIVERY MODE
			12/24/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Corrected Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--
 All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 10/28/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-113. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:
 a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☐ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	---

/CASSANDRA F COX/
 Primary Examiner, Art Unit 2849

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) _____		21 December 2020 O.G. Print Claim(s) 30	O.G. Print Figure 5A

<i>Issue Classification</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08

NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)		113	
(Date)			
/CASSANDRA F COX/		21 December 2020	
Primary Examiner, Art Unit 2849		O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)		30	5A
(Date)			

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	17	16	41	31	26	46	61	61	76	76	91	91	106	106
3	2	27	17	42	32	48	47	62	62	77	77	92	92	107	107
4	3	28	18	43	33	2	48	63	63	78	78	93	93	108	108
5	4	29	19	44	34	49	49	64	64	79	79	94	94	109	109
6	5	30	20	45	35	50	50	65	65	80	80	95	95	110	110
7	6	31	21	18	36	51	51	66	66	81	81	96	96	111	111
8	7	32	22	19	37	52	52	67	67	82	82	97	97	112	112
9	8	33	23	20	38	53	53	68	68	83	83	98	98	113	113
10	9	34	24	21	39	54	54	69	69	84	84	99	99		
11	10	35	25	22	40	55	55	70	70	85	85	100	100		
12	11	36	26	23	41	56	56	71	71	86	86	101	101		
13	12	37	27	24	42	57	57	72	72	87	87	102	102		
14	13	38	28	25	43	58	58	73	73	88	88	103	103		
15	14	39	29	46	44	59	59	74	74	89	89	104	104		
16	15	40	30	47	45	60	60	75	75	90	90	105	105		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 21 December 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

PRINTER RUSH
(PTO ASSISTANCE)

Application: 16218418

Examiner: Cox

GAU: 2849

From:

Location: FDC

Creation Date: 12/15/2020

Week Date: 8/17/2020

<u>DOC CODE</u>	<u>DOC DATE</u>	<u>MISCELLANEOUS</u>
☐ 1449		☐ Continuing Data
☐ IDS		☐ Foreign Priority
☒ CLM	<u>09/25/2020</u>	☐ Document Legibility
☒ IIFW/FWCLM	<u>11/25/2020</u>	☐ Fees
☐ SRFW		☐ Petition (TC)
☐ DRW		☐ Other
☐ OATH		
☐ 312		
☐ SPEC		

[RUSH] Message:

Please correct the claim dependencies

Per the set of claims dated 09/25/2020 and the most recent claim index (11/25/2020), original claims 2, 7, 8 and 10 - 14 all now depend on New claim 48. A claim can not depend on a subsequent numbered claim.

Thank You

[XRUSH] Response:

A new IIFW has been mailed correcting the claim numbering.

Initials: /C.F.C/

To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Dec 24, 2020 03:57:59 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	NOA	12/24/2020	00614-0183001

To view your correspondence online or update your email addresses, please visit us anytime at <https://portal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

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United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	01/26/2021	10903734	00614-0183001	1092

26214 7590 01/06/2021
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 58 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Vicor Corporation, Andover, MA;
Patrizio Vinciarelli, Boston, MA;

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To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Jan 07, 2021 04:10:52 AM

Dear PAIR Customer:

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P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	ISSUE.NTF	01/06/2021	00614-0183001

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Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor	:	Patrizio Vinciarelli	Art Unit	:	2849
Patent No.	:	10,903,734	Examiner	:	Cassandra F. Cox
Issue Date	:	January 26, 2021	Conf. No.	:	1092
Serial No.	:	16/218,418			
Filed	:	December 12, 2018			
Title	:	DELIVERING POWER TO SEMICONDUCTOR LOADS			

Attention Certificate of Corrections Branch
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REQUEST FOR CERTIFICATE OF CORRECTION

Please issue a certificate of correction in accordance with the attached form.

One or more of the errors to be corrected were made by the applicant. The fee of \$160 is being paid with this request.

Apply any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: April 20, 2021 _____

/Rex I. Huang/ _____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

**UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION**

Page 1 of 1

PATENT NO. : 10,903,734
APPLICATION NO.: 16/218,418
ISSUE DATE : January 26, 2021
INVENTOR(S) : Patrizio Vinciarelli

It is certified that errors appear in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Page 3, col. 1, Other Publications, line 35, delete “Intersil” and insert - - Intersil - -.

Sheet 11 of 13, FIG. 9, middle of figure, delete “Panel Sandwich Processes” and insert - - Panel Sandwich Processes - -.

Sheet 11 of 13, FIG. 9, block 624, delete “Singulate Sandwich Structure” and insert - - Singulate Sandwich Structure - -.

Col. 2, line 47, delete “TIN” and insert - - IIN - -.

Col. 9, line 23, delete “terminals” and insert - - Terminals - -.

Col. 19, line 14, in claim 9, delete “TIN” and insert - - IIN - -.

Col. 24, line 33, in claim 46, delete “claim 44” and insert - - claim 45 - -.

MAILING ADDRESS OF SENDER (Please do not use customer number below):

Fish & Richardson P.C.
P.O. Box 1022
Minneapolis, MN 55440-1022

This collection of information is required by 37 CFR 1.322, 1.323, and 1.324. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1.0 hour to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Attention Certificate of Corrections Branch, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and selection option2.

Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
CERTIFICATE OF CORRECTION	1811	1	160	160

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	42496637
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	20-APR-2021
Filing Date:	12-DEC-2018
Time Stamp:	10:55:56
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 160
RAM confirmation Number	E20214JA56121379
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

--

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Certificate of Correction	00614_0183001_Trans_COC.pdf	180995	no	2
			5d999abd9ee9f384eecd99df6444d3445f014ea8		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30227	no	2
			1bf941705494150046b91f57dec73c406a5ebd43		

Warnings:

Information:

Total Files Size (in bytes):			211222
------------------------------	--	--	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	05/07/2021		
FISH & RICHARDSON P.C. (VLT, INC)			EXAMINER	
P.O. BOX 1022			COX, CASSANDRA F	
MINNEAPOLIS, MN 55440-1022				
			ART UNIT	PAPER NUMBER
			2849	
			NOTIFICATION DATE	DELIVERY MODE
			05/07/2021	ELECTRONIC

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Patent No.: 10903734
Issue Date: 26 January 2021
Appl. No.: 16/218,418
Filed: 12 December 2018

PART (A) RESPONSE FOR CERTIFICATES OF CORRECTION

This is a decision on the Certificate of Correction request filed ____.

The request for issuance of Certificate of Correction for the above-identified correction(s) under the provisions of 37 CFR 1.322 and/or 1.323 is hereby:

(Check one)

☒ Approved ☐ Approved in Part ☐ Denied

Comments: ____

PART (B) PETITION UNDER 37 CFR 1.324 OR 37 CFR 1.48

☐ This is a decision on the petition filed ____ to correct inventorship under 37 CFR 1.324.

☐ This is a decision on the request under 37 CFR 1.48, petition filed _____. In view of the fact that the patent has already issued, the request under 37 CFR 1.48 has been treated as a petition to correct inventorship under 37 CFR 1.324.

The petition is hereby: ☐ Granted ☐ Dismissed

Comment: ____

The patented filed is being forwarded to Certificate of Corrections Branch for issuance of a certificate naming only the actual inventor or inventors.

/LINCOLN D DONOVAN/
Supervisory Patent Examiner, Art Unit 2842
Technology Center 2800
Phone: (571)272-1988

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 10,903,734 B1
APPLICATION NO. : 16/218418
DATED : January 26, 2021
INVENTOR(S) : Patrizio Vinciarelli

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

On the Title Page

Page 3, Column 1, Item (56) Other Publications, Line 35, delete "Intersil" and insert -- Intersil --.

In the Drawings

Sheet 11 of 13, FIG. 9, middle of figure, delete "Panel Sandwhich Processes" and insert -- Panel Sandwich Processes --.

Sheet 11 of 13, FIG. 9, block 624, delete "Singulate Sandwhich Structure" and insert -- Singulate Sandwich Structure --.

In the Specification

Column 2, Line 47, delete "TIN" and insert -- IIN --.

Column 9, Line 23, delete "terminals" and insert -- Terminals --.

In the Claims

Column 19, Line 14, in Claim 9, delete "TIN" and insert -- IIN --.

Column 24, Line 33, in Claim 46, delete "claim 44" and insert -- claim 45 --.

Signed and Sealed this
First Day of June, 2021



Drew Hirshfeld
*Performing the Functions and Duties of the
Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office*

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor	:	Patrizio Vinciarelli	Art Unit	:	2849
Patent No.	:	10,903,734	Examiner	:	Cassandra F. Cox
Issue Date	:	January 26, 2021	Conf. No.	:	1092
Serial No.	:	16/218,418			
Filed	:	December 12, 2018			
Title	:	DELIVERING POWER TO SEMICONDUCTOR LOADS			

Attention Certificate of Corrections Branch
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REQUEST FOR CERTIFICATE OF CORRECTION

Please issue a certificate of correction in accordance with the attached form.

All errors to be corrected were made in printing by the Patent and Trademark Office, and no fee is believed to be due.

Apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: March 15, 2024_____

/Rex I. Huang/
Rex I. Huang, Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

**UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION**

Page 1 of 1

PATENT NO. : 10,903,734
APPLICATION NO.: 16/218,418
ISSUE DATE : January 26, 2021
INVENTOR(S) : Patrizio Vinciarelli

It is certified that errors appear in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Col. 29, line 29, in Claim 100, delete “claim 32, 49” and insert - - claim 42, 49 - -.

Col. 30, line 13, Claim 107, delete “claim 30, 32, 33, 49” and insert - - claim 40, 42, 43, 49 - -.

Col. 30, line 16, Claim 108, delete “claim 30, 32, 33, 49” and insert - - claim 40, 42, 43, 49 - -.

Col. 30, line 31, Claim 111, delete “claim 30, 32, 33, 49” and insert - - claim 40, 42, 43, 49 - -.

Col. 30, line 34, Claim 112, delete “claim 32, 49” and insert - - claim 42, 49 - -.

Col. 30, line 38, Claim 113, delete “claim 30, 32, 33, 49” and insert - - claim 40, 42, 43, 49 - -.

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
16/218,418RECEIPT DATE / TIME
03/15/2024 09:58:48 AM Z ETATTORNEY DOCKET #
00614-0183001

Title of Invention

DELIVERING POWER TO SEMICONDUCTOR LOADS

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # 10903734

CONFIRMATION # 1092

FILED BY Rita Liston

PATENT CENTER # 64697463

FILING DATE 12/12/2018

CUSTOMER # 26214

FIRST NAMED
INVENTOR Patrizio VinciarelliCORRESPONDENCE
ADDRESS -

AUTHORIZED BY Rex Huang

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00614_0183001_Trans_COC 031524.pdf	2	Request for Certificate of Correction	176 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00614_0183001_Trans_COC 031524.pdf	E6332BD9D550F1243535DC690D30A39DA4AA83F09CAFB051 D4C430A4D9F4F7DE4F1B4D81A7E8DD3A669BD1399867D97A 288DA2C8176B04A12507FDD080D2FD7F

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214 7590 04/09/2024 FISH & RICHARDSON P.C. (VLT, INC) P.O. BOX 1022 MINNEAPOLIS, MN 55440-1022			EXAMINER COX, CASSANDRA F	
			ART UNIT	PAPER NUMBER
			2849	
			NOTIFICATION DATE	DELIVERY MODE
			04/09/2024	ELECTRONIC

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APPLICATION NO.	ISSUE DATE	PATENT NO.
16/218,418	09-Apr-2024	10903734

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

ELECTRONIC CERTIFICATE OF CORRECTION NOTIFICATION

Your electronic certificate of correction is now available, and can be accessed via Patent Center at <https://patentcenter.uspto.gov>.

The electronic certificate of correction is the official certificate of correction under 35 U.S.C. 254 or 255. For more information, please visit <https://www.uspto.gov/electronicgrants>.

TO: PATDOCTC@fr.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - eGrant Notification for Customer Number 26214

Tue Apr 09 06:02:03 EDT 2024

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Application	Patent	Document	Mailroom Date	Attorney Docket No.
16218418	10903734	ECOF.C.NTF	04/09/2024	00614-0183001

To examine your eGrant and eCofC online, please access [Patent Center](#) then search using the listed application or patent number. To download your eGrant, click on the eGrant and eCofC link on the left-hand menu or select the "Download eGrant" link on the Application data page. To download your eCofC, click on the eGrant and eCofC link on the left-hand menu.

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Sincerely,

The Patent Center Team

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 10,903,734 B1
APPLICATION NO. : 16/218418
DATED : January 26, 2021
INVENTOR(S) : Patrizio Vinciarelli

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 29, Line 29, Claim 100, delete “claim 32, 49” and insert -- claim 42, 49 --.

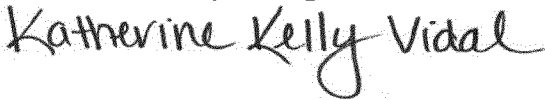
Column 30, Line 13, Claim 107, delete “claim 30, 32, 33, 49” and insert -- claim 40, 42, 43, 49 --.

Column 30, Line 16, Claim 108, delete “claim 30, 32, 33, 49” and insert -- claim 40, 42, 43, 49 --.

Column 30, Line 31, Claim 111, delete “claim 30, 32, 33, 49” and insert -- claim 40, 42, 43, 49 --.

Column 30, Line 34, Claim 112, delete “claim 32, 49” and insert -- claim 42, 49 --.

Column 30, Line 38, Claim 113, delete “claim 30, 32, 33, 49” and insert -- claim 40, 42, 43, 49 --.

Signed and Sealed this
Ninth Day of April, 2024

Katherine Kelly Vidal
Director of the United States Patent and Trademark Office

TO: PATDOCTC@fr.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 26214

Wed Apr 10 05:04:01 EDT 2024

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16218418	COCOUT	04/09/2024	00614-0183001

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